



Determinant

Department of Computer Engineering
Sharif University of Technology

Hamid R. Rabiee rabiee@sharif.edu

Maryam Ramezani maryam.ramezani@sharif.edu



Table of contents

01

Introduction

02

Bilinear Form:
Review and Continue

03

Multilinear
Form

04

Matrix
Determinant

05

Cramer's Rule

06

Linking Determinant to
other Course Concepts

07

Determinant
Properties



01

Introduction



Determinant of a matrix

The determinant of a 2×2 matrix $A = [a_{ij}]$ is the number:
Why???

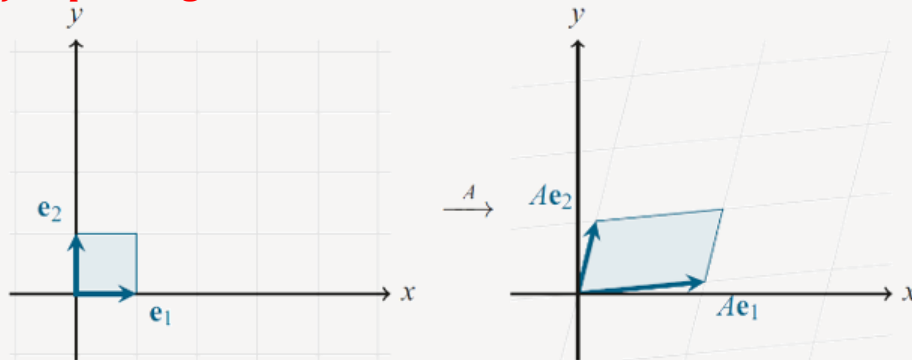
$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

□ The absolute value of the determinant of a matrix measures how much it expands space when acting as a linear transformation. That is, it is the area (or volume, or hypervolume, depending on the dimension) of the output of the unit square, cube, or hypercube after it is acted upon by the matrix.

Geometric interpretation

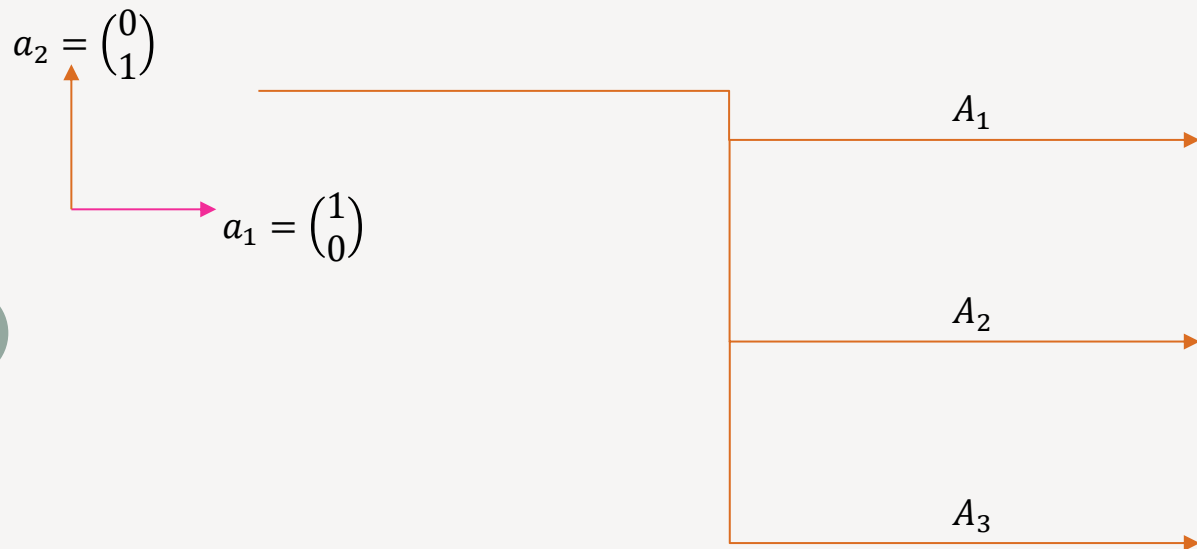
□ The volume is a n -alternating multilinear map on all n -parallelepipeds such that the volume of standard unit parallelepiped is one.

$$\frac{\text{volume of output region}}{\text{volume of input region}}$$



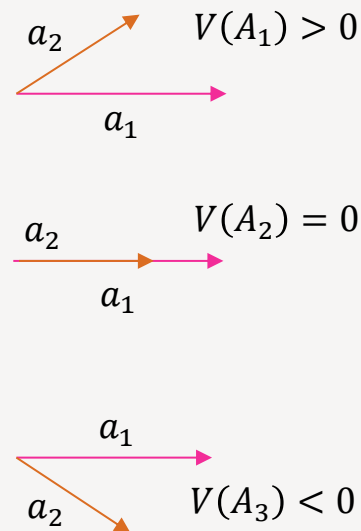
A 2×2 matrix A stretches the unit square (with sides e_1 and e_2) into a parallelogram with sides Ae_1 and Ae_2 (the columns of A). The determinant of A is the area of this parallelogram.

Geometric interpretation



$$A = [a_1 \quad a_2], a_2 = \alpha a_1$$

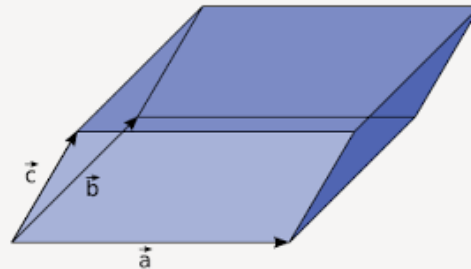
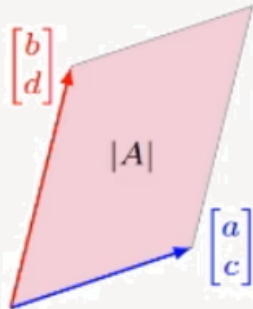
$$V(a_1, a_2) = -V(a_2, a_1)$$



Determinants as Area or Volume

- If A is a 2×2 matrix, the **area** of the parallelogram determined by the columns of A is $\det(A)$
- If A is a 3×3 matrix, the **volume** of the parallelepiped determined by the columns of A is $\det(A)$
- Examples:

Volume of $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ It is a rotation with θ degree



Volume

Definition

Every n -dimensional parallelepiped with $\{a_1, \dots, a_n\}$ as legs is associated with a real number, called its volume which has the following properties:

If we stretch a parallelepiped by multiplying one of its legs by a scalar λ , its volume gets multiplied by λ .

If we add a vector ω to i -th legs of a n -dimensional parallelepiped with $\{a_1, \dots, a_i, a_{i+1}, \dots, a_n\}$, then its volume is the sum of the volume from $\{a_1, \dots, a_{i-1}, a_i, a_{i+1}, \dots, a_n\}$ and the volume of $\{a_1, \dots, a_{i-1}, \omega, a_{i+1}, \dots, a_n\}$.

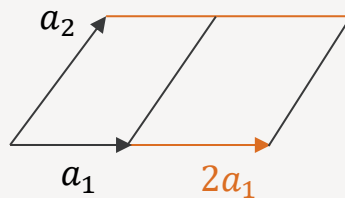
The volume changes sign when two legs are exchanged.

The volume of the parallelepiped with $\{e_1, \dots, e_n\}$ is one.

$$\phi : \underbrace{V \times \dots \times V}_n \rightarrow \mathbb{R}$$

Volume Example

- 2D Case
 - $V(a_1, a_2)$
 - $V(2 a_1, a_2) = 2V(a_1, a_2)$
 - $V(-a_1, a_2) = -V(a_1, a_2)$
 - $V(\beta a_2, a_1) = -\beta V(a_1, a_2)$



02

Bilinear Form: Review and Continue



Bilinear Form over a complex vector

Definition

Suppose V and W are vector spaces over the same field $\mathbb{C}$. Then a function $\alpha: V \times W \rightarrow \mathbb{C}$ is called a **bilinear form** if it satisfies the following properties:

It is **linear in its first argument**:

$$\alpha(\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}) = \alpha(\mathbf{v}_1, \mathbf{w}) + \alpha(\mathbf{v}_2, \mathbf{w}) \text{ and}$$

$$\alpha(\lambda \mathbf{v}_1, \mathbf{w}) = \lambda \alpha(\mathbf{v}_1, \mathbf{w}) \text{ for all } \lambda \in \mathbb{C}, \mathbf{v}_1, \mathbf{v}_2 \in V, \text{ and } \mathbf{w} \in W.$$

It is **conjugate linear in its second argument**:

$$\alpha(\mathbf{v}, \mathbf{w}_1 + \mathbf{w}_2) = \alpha(\mathbf{v}, \mathbf{w}_1) + \alpha(\mathbf{v}, \mathbf{w}_2) \text{ and}$$

$$\alpha(\mathbf{v}, \lambda \mathbf{w}_1) = \bar{\lambda} \alpha(\mathbf{v}, \mathbf{w}_1) \text{ for all } \lambda \in \mathbb{C}, \mathbf{v} \in V, \text{ and } \mathbf{w}_1, \mathbf{w}_2 \in W.$$

The set of bilinear forms on $\mathbf{v}$ is denoted by $\mathbf{v}^2$.

Alternating bilinear form

Definition

A bilinear form $\alpha \in V^{(2)}$ is called alternating if

$$\alpha(v, v) = 0$$

for all $v \in V$. The set of alternating bilinear forms on V is denoted by $V_{alt}^{(2)}$.

Example

Suppose $\varphi, \tau \in V'$. Then the bilinear form α on V defined by is alternating.

$$\alpha(u, \omega) = \varphi(u)\tau(\omega) - \varphi(\omega)\tau(u)$$

Alternating bilinear form

Theorem (1)

A bilinear form α on V is alternating if and only if

$$\alpha(u, \omega) = -\alpha(\omega, u)$$

For all $u, \omega \in V$.



Proof

Alternating bilinear form

Theorem (2)

The sets $V_{sym}^{(2)}$ and $V_{alt}^{(2)}$ are subspaces of $V^{(2)}$. Furthermore,

$$V^{(2)} = V_{sym}^{(2)} \oplus V_{alt}^{(2)}$$



Proof

03

Multilinear Form



Multilinear Forms

Definition

Suppose $\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_p$ are vector spaces over the same field $\mathbb{F}$. A function

$$f : \mathcal{V}_1 \times \mathcal{V}_2 \times \dots \times \mathcal{V}_p \rightarrow \mathbb{F}$$

is called a multilinear form ($f \in V^{(p)}$) if, for each $1 \leq j \leq p$ and each $v_1 \in \mathcal{V}_1, v_2 \in \mathcal{V}_2, \dots, v_p \in \mathcal{V}_p$, it is the case that the function $g : \mathcal{V}_j \rightarrow \mathbb{F}$ defined by

$$g(v) = f(v_1, \dots, v_{j-1}, v, v_{j+1}, \dots, v_p) \quad \text{for all } v \in \mathcal{V}_j$$

is a linear form.

Example

Suppose $\alpha, \rho \in V^{(2)}$. Define a function $\beta: V^4 \rightarrow \mathbb{F}$ by then $\beta \in V^{(4)}$ (which means multilinear)

$$\beta(v_1, v_2, v_3, v_4) = \alpha(v_1, v_2)\rho(v_3, v_4)$$

Multilinear Forms

Definition

Suppose m is a positive integer.

- An m -linear form α on V is called *alternating* if $\alpha(v_1, \dots, v_m) = 0$ whenever $v_1, \dots, v_m$ is a list of vectors in V with $v_j = v_k$ for some two distinct values of j and k in $\{1, \dots, m\}$.
- $V_{alt}^{(m)} = \{\alpha \in V^{(m)} : \alpha \text{ is an alternating } m\text{-linear form on } V\}$.

Theorem (3)

$V_{alt}^{(m)}$ is a subspace of $V^{(m)}$.

Proof

Review: Characterization of Linearly Dependent sets

Theorem (4)

An indexed set $S = \{v_1, \dots, v_n\}$ of two or more vectors is linearly dependent **if and only if** at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then **some** v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \dots, v_{j-1}$.

☐ Does not say that every vector

☐ Does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Alternating multilinear forms and linear dependence

Theorem (5)

Suppose m is a positive integer and α is an alternating m -linear form on V . If $v_1, \dots, v_m$ is a linearly dependent list in V , then

$$\alpha(v_1, \dots, v_m) = 0$$

Proof

No nonzero alternating m -linear forms for $m > \dim V$

Theorem (6)

Suppose $m(\text{number of vectors}) > \dim V$. Then there is an alternating m -linear form on V .



Proof



Swapping input vectors in an alternating multilinear form

Theorem (7)

Suppose m is a positive integer, α is an alternating m -linear form on V , and $v_1, \dots, v_m$ is a list of vectors in V . Then swapping the vectors in any switch of $\alpha(v_1, \dots, v_m)$ changes the value of α by a factor of -1 .

Okey, clearing up the last detail. Suppose we know that $A(e_1, e_2, e_3, e_4, e_5) = 7$. What should $A(e_3, e_5, e_1, e_2, e_4)$ be?

$$\begin{aligned} A(e_3, e_5, e_1, e_2, e_4) &= -A(e_3, e_4, e_1, e_2, e_5) \\ &= A(e_3, e_2, e_1, e_4, e_5) \\ &= -A(e_1, e_2, e_3, e_4, e_5) = -7 \end{aligned}$$

What if we did the switching in a different order? Would we get the same sign? It turns out that, yes, we would!

Permutation

Definition

Suppose m is a positive integer.

A permutation of $(1, \dots, m)$ is a list $(j_1, \dots, j_m)$ that contains each of the numbers $1, \dots, m$ exactly once.

The set of all permutations of $(1, \dots, m)$ is denoted by $\text{perm } m$.

Example

What we need to show is that there is a way to assign a sign to

- every permutation of $\{1, 2, 3, \dots, k\}$ such that, switching the order of any two elements, switches the sign. For example:

$$(1, 2, 3) \rightsquigarrow 1 \quad (1, 3, 2) \rightsquigarrow -1$$

$$(2, 1, 3) \rightsquigarrow -1 \quad (2, 3, 1) \rightsquigarrow 1$$

$$(3, 1, 2) \rightsquigarrow 1 \quad (3, 2, 1) \rightsquigarrow -1$$

Here is the rule: The sign of $(\sigma(1), \sigma(2), \dots, \sigma(k))$ is

$$(-1)^{\#\{(i,j) : i < j \text{ and } \sigma(i) > \sigma(j)\}}.$$

$$A(e_{j_1}, e_{j_2}, \dots, e_{j_k}) = \text{sign}(\sigma) A(e_{j_{\sigma(1)}}, e_{j_{\sigma(2)}}, \dots, e_{j_{\sigma(k)}}).$$

Permutation

Definition

The *sign* of a permutation $(j_1, \dots, j_m)$ is defined by

$$\text{sign}(j_1, \dots, j_m) = (-1)^N$$

Where N is the number of pairs of integers (k, l) with $1 \leq k < l \leq m$ such that k appears after l in the list $(j_1, \dots, j_m)$.

Example

The permutation $(1, \dots, m)$ [no changes in the natural order] has sign 1.

The only pair of integers (k, l) with $k < l$ such that k appears after l in the list $(2, 1, 3, 4)$ is $(1, 2)$. Thus the permutation $(2, 1, 3, 4)$ has sign -1 .

In the permutation $(2, 3, \dots, m, 1)$, the only pairs (k, l) with $k < l$ that appear with changed order are $(1, 2), (1, 3), \dots, (1, m)$. Because we have $m - 1$ such pairs, the sign of this permutation equals $(-1)^{m-1}$.

Permutations and alternating multilinear forms

Theorem (8)

Suppose m is a positive integer and $\alpha \in V_{alt}^{(m)}$. Then

$$\alpha(v_{j_1}, \dots, v_{j_m}) = \text{sign}(j_1, \dots, j_m) \alpha(v_1, \dots, v_m)$$

for every list $v_1, \dots, v_m$ of vectors in V and all $(j_1, \dots, j_m) \in \text{perm } m$.

Proof

Formula for $(\dim V)$ -linear alternating forms on V

Theorem (9)

Let $n = \dim V$. Suppose $e_1, \dots, e_n$ is a basis of V and $v_1, \dots, v_n \in V$. For each $k \in \{1, \dots, n\}$, let $b_{1,k}, \dots, b_{n,k} \in \mathbf{F}$ be such that

$$v_k = \sum_{j=1}^n b_{j,k} e_j$$

Then

$$\alpha(v_1, \dots, v_n) = \alpha(e_1, \dots, e_n) \sum_{(j_1, \dots, j_n) \in \text{perm}(n)} (\text{sign}(j_1, \dots, j_n)) b_{j_1, 1} \dots b_{j_n, n}$$

for every alternating n -linear form α on V .

Proof

04

Matrix Determinant



Definition

- In **Theorem (9)** if we consider $\alpha(e_1, \dots, e_n) = 1$, then we can say that determinant is a multilinear alternating form (volume of the transformed vectors.)
- **Determinant is defined for square matrices.**

Definition

Suppose that n is a positive integer and A is an n -by- n square matrix. Then:

$$\det A = \sum_{(j_1, \dots, j_n) \in \text{perm}(n)} (\text{sign}(j_1, \dots, j_n)) A_{j_1, 1} \dots A_{j_n, n}$$

Definition

Note

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\}$$

If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\}$$

Example

Example

Determinant of 2*2 matrix

Determinant of 3*3 matrix

$$V\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}\right) = V\left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix}\right)$$

$$V\left(a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}, c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$a V\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) + b V\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$ac V\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + ad V\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) + bc V\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + bd V\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) + 0 + ad V\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) + bc V\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + 0$$

$$= ad - bc V\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = ad - bc$$

$$\Rightarrow V\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}\right) = ad - bc$$

Famous Formula of Determinant

Determinant

Example

Let $n = \dim V$.

- If I is the identity operator on V , then $\alpha_1 = \alpha$ for all $\alpha \in V_{alt}^{(n)}$. Thus, $\det I = 1$.
- More generally, if $\lambda \in \mathbf{F}$, then $\alpha_{\lambda I} = \lambda^n \alpha$ for all $\alpha \in V_{alt}^{(n)}$. Thus, $\det(\lambda I) = \lambda^n$.
- Still more generally, if $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$, then $\alpha_{\lambda T} = \lambda^n \alpha_T = \lambda^n (\det T) \alpha$ for all $\alpha \in V_{alt}^{(n)}$. Thus, $\det(\lambda T) = \lambda^n \det T$.

Definition of Submatrix A_{ij}

Definition

For any square matrix A , let A_{ij} denote the submatrix formed by deleting the i th row and j th column of A



For instance, if

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

A_{12} is

$$A_{12} = \begin{bmatrix} 2 & 4 & -1 \\ 3 & 0 & 7 \\ 0 & -2 & 0 \end{bmatrix}$$



Recursive Definition of Determinant

Definition

The determinant of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det(A_{1j})$, with **plus and minus signs alternating**, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\begin{aligned}\det(A) &= a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{1+n} a_{1n} \det(A_{1n}) \\ &= \sum_{j=1}^n a_{1j} \underbrace{(-1)^{1+j}}_{C_{1j}} \det(A_{1j})\end{aligned}$$

Recursive Definition of Determinant

□ 2×2 matrix

$$|A| = \sum_{j=1}^n (-1)^{i+j} a_{ij} |A_{ij}| \quad i = 1$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow |A| = (-1)^{1+1} a_{11} |A_{11}| + (-1)^{1+2} a_{12} |A_{12}|$$

$$= a \begin{vmatrix} \square & \square \\ \square & d \end{vmatrix} - b \begin{vmatrix} \square & \square \\ c & \square \end{vmatrix}$$

$$= ad - bc$$

Example

$$\begin{vmatrix} -1 & 2 \\ -3 & 1 \end{vmatrix} = (-1) \times (1) - (2) \times (-3) = 5$$

Recursive Definition of Determinant

□ 3×3 matrix $|A| = \sum_{j=1}^n (-1)^{i+j} a_{ij} |A_{ij}| \quad i = 1$


$$\begin{aligned} A &= \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow |A| \\ &= (-1)^{1+1} a_{11} |A_{11}| + (-1)^{1+2} a_{12} |A_{12}| + (-1)^{1+3} a_{13} |A_{13}| \\ &= a \begin{vmatrix} \square & \square & \square \\ \square & e & f \\ \square & h & i \end{vmatrix} - b \begin{vmatrix} \square & \square & \square \\ d & \square & f \\ g & \square & i \end{vmatrix} + c \begin{vmatrix} \square & \square & \square \\ d & e & \square \\ g & h & \square \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei + bfg + cdh - afh - bdi - ceg \end{aligned}$$


Cofactor

Definition

Given $A = [a_{ij}]$, the (i, j) -cofactor of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det(A_{ij})$$

Then

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

Which is a cofactor expansion across the first row of A .

Cofactor Expansion

Important

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion **across any row or down any column**. The expansion across the i th row using the cofactor is

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

The cofactor expansion down the j th column is

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

Cofactor Expansion

Example

$$A = \begin{bmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 5 & 4 \\ 5 & 3 & -1 \end{bmatrix}$$

$$|A| = +1 \times \begin{vmatrix} 5 & 4 \\ 3 & -1 \end{vmatrix} - 0 \times \begin{vmatrix} 2 & 4 \\ 5 & -1 \end{vmatrix} + 1 \times \begin{vmatrix} 2 & 5 \\ 5 & 3 \end{vmatrix} = -36$$

$$|A| = -0 \times \begin{vmatrix} 2 & 4 \\ 5 & -1 \end{vmatrix} + 5 \times \begin{vmatrix} 1 & 1 \\ 5 & -1 \end{vmatrix} - 3 \times \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} = -36$$

05

Cramer's Rule



Cramer's Rule

□ $Ax = b$ and A is invertible

$$A = [A_1 \quad \dots \quad A_n] \quad I$$

$$= [e_1 \quad \dots \quad e_n]$$

$$AI = A \Rightarrow A[e_1 \quad \dots \quad e_n] = [Ae_1 \quad \dots \quad Ae_n] = [A_1 \quad \dots \quad A_n]$$

$$A \overbrace{[e_1 \quad e_2 \quad \dots \quad x \quad \dots \quad e_n]}^{I_j(x)} = [Ae_1 \quad Ae_2 \quad \dots \quad Ax \quad \dots \quad Ae_n]$$

$$= \underbrace{[A_1 \quad A_2 \quad \dots \quad b \quad \dots \quad A_n]}_{A_j(b)}$$

$$|I_2(x)| = \begin{vmatrix} 1 & x_1 & 0 \\ 0 & x_2 & 0 \\ 0 & x_3 & 1 \end{vmatrix} = x_2 \Rightarrow |I_j(x)| = x_j$$

$$AI_j(x) = A_j(b) \Rightarrow |A||I_j(x)| = |A_j(b)| \Rightarrow x_j = \frac{|A_j(b)|}{|A|}$$

Cramer's Rule

Note

□ Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

□ $x_i = \frac{|A_i(\mathbf{b})|}{|A|}, \quad i = 1, 2, \dots, n$

Example

$$\begin{cases} x_1 - x_2 + 2x_3 = 1 \\ x_1 + x_2 - x_3 = 2 \\ 2x_1 - 3x_2 + x_3 = -1 \end{cases}$$

$$\Rightarrow x_2 = \frac{\begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & -1 & 2 \\ 1 & 1 & -1 \\ 2 & -3 & 1 \end{vmatrix}} = \frac{-12}{-3} = 4$$

A Formula for A^{-1}

The j -th column of A^{-1} is a vector x that satisfies $Ax = e_j$

By Cramer's rule $\{(i, j) - \text{entry of } A^{-1}\} = x_i = \frac{|A_i(e_j)|}{|A|}$

$$|A_i(e_j)| = (-1)^{i+j} |A_{ji}|$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

The matrix of cofactors is called the **adjugate** (or **classical adjoint**) of A , denoted by $\text{adj } A$.

$$\begin{aligned} A &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ [C_{ij}] &= \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ \Rightarrow A^{-1} &= \frac{1}{|A|} [C_{ij}]^T = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \end{aligned}$$

A Formula for A^{-1}

Important

Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{|A|} \operatorname{adj} A$$

06

Linking Determinant to other Course Concepts



Gaussian Elimination & Determinant

Note

- Row operations
- Let A be a square matrix.
- If a multiple of one row of A is added to another row to produce a matrix B , then $\det(B) = \det(A)$
- If two rows of A are interchanged to produce B , then $\det(B) = -\det(A)$
- If one row of A is multiplied by k to produce B , then $\det(B) = k \cdot \det(A)$



Matrices Multiplication

Theorem (10)

if A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \det(B)$



Transformations

Theorem (11)

○ Show that the determinant, $\det: \mathcal{M}_n(\mathbb{F}) \rightarrow \mathbb{F}$ is not a linear transformation when $n \geq 2$

In general, $\det(A + B) \neq \det(A) + \det(B)$

Matrix Inverse

Theorem (12)

A square matrix A is invertible if and only if $\det(A) \neq 0$

Theorem (13)

The determinant of the inverse of an invertible matrix is the inverse of the determinant.

Proof

$$AA^{-1} = I \Rightarrow |AA^{-1}| = |I| = 1 \Rightarrow |A||A^{-1}| = 1 \Rightarrow |A^{-1}| = |A|^{-1}$$

Matrix Determinant

Example

□ Compute $\det(A)$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

□ The determinant of orthogonal matrix is ...

QR Decomposition

Note

□ If $A \in \mathcal{M}_n$ has QR decomposition $A = UT$ with $U \in \mathcal{M}_n$ unitary and $T \in \mathcal{M}_n$ upper triangular, then

$$|\det(A)| = t_{1,1} \cdot t_{2,2} \dots t_{n,n}.$$

Example

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 2 & -2 \\ -2 & 2 & 1 \end{bmatrix} \text{ has QR decomposition } A = UT \text{ with } U = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \text{ and } T = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

07

Determinant Properties



Good News!

If you understand the properties of determinant as an alternating multilinear form, you can skip the following slides, because you know all of them!! 😁



Properties

- (1) If one row or column is zero, then determinant is zero


$$\begin{vmatrix} 0 & 0 & 0 \\ a & b & c \\ d & e & f \end{vmatrix} = 0$$

- Determinant of zero matrix is...

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j})$$
$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)}$$


Properties

- (2) If two rows or columns of matrix are same, then determinant is zero.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & -2 & 3 \\ 5 & 3 & -1 \end{bmatrix}$$

$$|A| = +1 \times \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} - (-2) \times \begin{vmatrix} 1 & 3 \\ 5 & -1 \end{vmatrix} + 3 \times \begin{vmatrix} 1 & -2 \\ 5 & 3 \end{vmatrix}$$

$$|A| = -1 \times \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} + (-2) \times \begin{vmatrix} 1 & 3 \\ 5 & -1 \end{vmatrix} - 3 \times \begin{vmatrix} 1 & -2 \\ 5 & 3 \end{vmatrix}$$

Properties

- (3) If two rows or columns of matrix are interchanged, the sign of determinant is changes!
- (4) $\det(I) = 1$



Properties

□ (5) Row and Column Operations

- If a multiple of one row/column of A is added to another row/column to produce a matrix B , then $\det(A) = \det(B)$.

Proof?

Example

$$\begin{vmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 0 & 0 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 2 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 6 \\ 1 & 1 & 3 \\ 1 & -1 & 4 \end{vmatrix}$$

Properties

- (6) If A is a triangular matrix, then $\det(A)$ is the product of the entries on the main diagonal of A .



$$\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

$$\begin{vmatrix} a & 0 & 0 \\ d & b & 0 \\ e & f & c \end{vmatrix} = abc$$

- Determinant of identity matrix is...

- U is unitary, so that $|\det(U)|=1$



Properties

- (7) If a column or row is multiply to k then determinant is multiply to k.


$$\begin{vmatrix} ka_{11} & \dots & ka_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix} = k \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}$$

- $|kA_{n \times n}| = k^n |A_{n \times n}|$



Properties

- (8) If a row/column is multiple of another row/column then determinant is



Properties

- (9) If columns/rows of matrix are linear dependent then its determinant is zero
- (10) If columns/rows of matrix are linear dependent if and only if its determinant is zero.
- (11) Transposing a matrix does not change the determinant.



Reference

- ❑ Chapter 3: Linear Algebra and Its Applications, David C. Lay.
- ❑ Chapter 9: Part B and C: Linear Algebra Done Right, Sheldon Axler.

